Query-Adaptive Diffusion Transformer for Efficient Any-Subset Conditional Sampling on Tabular Data and Beyond

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Abstract

Diffusion models provide a flexible framework for conditional generation across tabular and multimodal domains. Many scientific and statistical inference tasks, however, require arbitrary conditional sampling, such as estimating posterior distributions, conditional expectations, or uncertainty under any-subset of observed variables. Existing diffusion methods that support such flexibility typically denoise and sample over the full variable set, leading to computation that scales with data dimensionality rather than the query itself. We propose Query-Adaptive Diffusion Transformer (QuADiT), a diffusion framework that formulates conditional generation as query-specific inference. By representing variables as variable-length tokens, QuADiT restricts diffusion to only the conditioning and target variables specified by a query, eliminating redundant computation while preserving full conditional expressivity. We further study practical design and implementation choices, providing robust standard configurations. Finally, we demonstrate that the approach extends beyond tabular data to multimodal tasks, including tabular-conditioned image generation.

1. Introduction

Diffusion models provide a principled way to model complex joint distributions via iterative denoising (Ho et al., 2020; Song et al., 2021b). They gained widespread attention through strong image generation results, particularly in text-to-image settings with latent diffusion models (Rombach et al., 2022), and now appear across modalities, including tabular data such as TabDDPM (Kotelnikov et al., 2023), STaSy (Kim et al., 2023), and TabCSDI (Zheng & Charoenphakdee, 2022), speech synthesis such as DiffWave (Kong et al., 2021), and video generation such as Video Diffusion Models (Ho et al., 2022).

Many scientific and statistical tasks require any-subset conditional sampling, where a target subset of variables is generated given an arbitrary observed subset. Prior tabular diffusion methods such as TabDDPM, STaSy, and TabCSDI already support such flexibility through masking mechanisms. However, these methods denoise the full feature set regardless of the query and discard unneeded outputs, so compute and sampling cost scale with the total feature count rather than the query size. Moreover, various design choices for tabular diffusion, including prediction targets, noise schedules, and normalization, remain fragmented across prior work without a unified recommendation.

We propose Query-Adaptive Diffusion Transformer (QuADiT), a framework for efficient any-subset conditional sampling across tabular and multimodal data. Our focus is not on task-specific state-of-the-art performance but on a practical and reusable design. Our contributions are as follows.

- We introduce query-adaptive token processing that computes only the conditioning and target tokens specified by a query, reducing overall Transformer compute by avoiding irrelevant features.
- We distill existing tabular diffusion practices into a unified feature-token formulation within a DiT backbone and systematically explore key design choices, proposing a standard configuration with cosine schedule, v-prediction, and sandwich normalization.
- We show that the same framework extends to multimodal settings by treating image attributes in existing vision datasets as tabular features, enabling joint and conditional generation across tabular and image modalities within a single model.

Figure 1 illustrates a subset of tasks our framework supports, including marginal, joint, and conditional sampling across tabular and image modalities.

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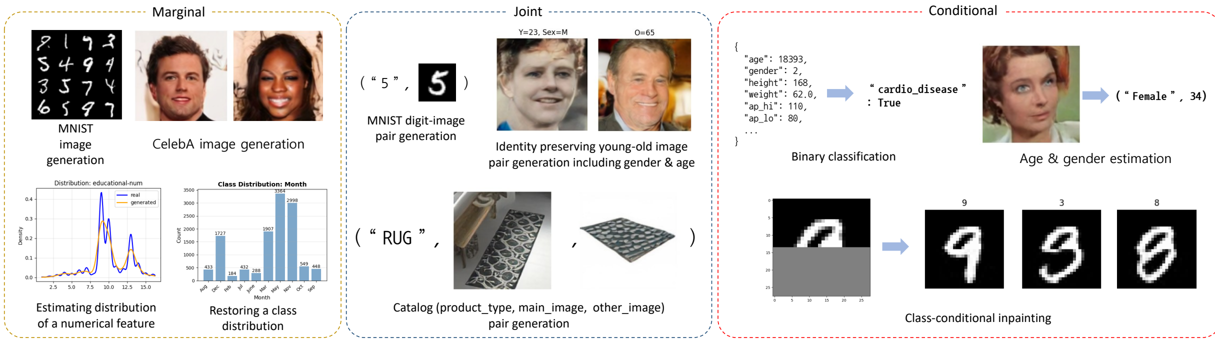

Figure 1. Overview of some tasks enabled by QuADiT: *marginal* generation (unconditional sampling of single features), *joint* generation (sampling multiple features together), and *conditional* generation (sampling target features given observed ones). Examples include image synthesis, distribution estimation, tabular-image pair generation, classification, regression, and inpainting. The framework is detailed in Section 4 and datasets are described in Section 5.

2. Background

We review diffusion foundations and recent advances toward generalized conditional sampling across modalities, focusing on broadly applicable backbones and conditioning strategies rather than task-specific architectures.

2.1. Diffusion Transformers

We denote a data sample by $x^{(0)} \in \mathbb{R}^d$ and define a forward noising chain with schedule $\{\beta_t\}_{t=1}^T$ as

$$q(x^{(t)}|x^{(t-1)}) = \mathcal{N}(\sqrt{1-\beta_t}\, x^{(t-1)}, \beta_t I). \tag{1}$$

This yields $q(x^{(t)}|x^{(0)}) = \mathcal{N}(\sqrt{\alpha_t}\, x^{(0)}, (1-\alpha_t)I)$ with $\alpha_t = \prod_{s=1}^{t}(1-\beta_s)$. The model learns the reverse process $p_\theta(x^{(t-1)}|x^{(t)}, c)$ and is commonly trained to predict noise via $\|\epsilon - \epsilon_\theta(x^{(t)}, t, c)\|^2$, where c denotes conditioning. We write diffusion timesteps as superscripts and feature indices as subscripts. Early diffusion models relied on U-Net backbones, but Transformer denoisers now match or surpass them. U-ViT (Bao et al., 2023a) replaces the U-Net with a ViT-style backbone, treating time, conditions, and noisy patches as tokens. Diffusion Transformer (DiT) (Peebles & Xie, 2023) scales this idea with latent patches and adaptive layer normalization (adaLN-Zero) for conditioning and shows strong compute–performance scaling. Tokenization also supports heterogeneous data types (Gorishniy et al., 2021), which we leverage for tabular settings.

2.2. Discrete Diffusion for Categorical Variables

Discrete diffusion was introduced for binary variables in the original diffusion probabilistic model (Sohl-Dickstein et al., 2015), which specifies forward and reverse kernels for a Bernoulli chain. For categorical variables, argmax flows and multinomial diffusion (Hoogeboom et al., 2021) formalize one-hot representations with argmax decoding. For a categorical variable $y \in \{1, \ldots, C\}$, we use a one-hot vector $z = e_y$ and decode with $\hat{y} = \arg\max_k z_k$, while the reverse model outputs class probabilities $\pi_\theta(x^{(t)}, t) \in \Delta^{C-1}$. We follow this representation for categorical columns.

Beyond uniform categorical corruption, D3PMs (Austin et al., 2021) generalize multinomial diffusion with structured transition matrices, and VQ-Diffusion (Gu et al., 2022) uses mask-and-replace transitions on discrete codebook tokens.

2.3. Tabular Diffusion Models

For mixed-type tables, let $x = (x_{\text{num}}, x_{\text{cat}})$ and let z be the one-hot representation of a categorical field with C classes. A common formulation applies Gaussian noising to numerical features and multinomial noising to categorical ones.

$$\begin{aligned} q(x_{\text{num}}^{(t)}|x_{\text{num}}^{(t-1)}) &= \mathcal{N}(\sqrt{1-\beta_t}\, x_{\text{num}}^{(t-1)}, \beta_t I), \\ q(z^{(t)}|z^{(t-1)}) &= \text{Cat}\big((1-\beta_t)z^{(t-1)} + \beta_t C^{-1}\mathbf{1}\big). \end{aligned} \tag{2}$$

This aligns with feature-wise tokenization and per-feature projections used in practice.

Building on this, TabDDPM (Kotelnikov et al., 2023) combines Gaussian diffusion for numerical features with multinomial diffusion for categorical ones. STaSy (Kim et al., 2023) applies score-based modeling with self-paced training, while CoDi (Lee et al., 2023) co-evolves continuous and discrete diffusion models that condition on each other to preserve inter-feature dependencies. For missing-value imputation, TabCSDI (Zheng & Charoenphakdee, 2022) adapts conditional score-based diffusion and compares one-hot, analog-bits, and feature-tokenization schemes for categorical variables.

Tabular Transformers standardize heterogeneous fields into a shared token space. Feature tokenization (Gorishniy et al., 2021) assigns each column a token embedding via per-feature linear projections for numeric values and embeddings for categorical values, enabling uniform self-attention across columns and informing feature-level conditioning and masking.

3. Conditioning Paradigms

We now turn to conditioning paradigms and focus on mechanisms that transfer across modalities, since mask design and token-level conditioning largely determine whether any-subset queries can be supported efficiently.

3.1. Conditioned Image Generation

Conditional diffusion in images began with class-conditional generation. Conditioning was injected through feature-wise modulation in U-Net blocks, and classifier guidance (Dhariwal & Nichol, 2021) used an external classifier to steer samples toward the desired class. Classifier-free guidance (CFG) (Ho & Salimans, 2022) later replaced the external classifier by mixing conditional and unconditional scores and became the default for high-fidelity conditional synthesis.

Text conditioning expanded the representation space from discrete labels to rich prompts, with cross-attention as the standard mechanism to inject token-level semantics, as in GLIDE (Nichol et al., 2022) and latent diffusion.

Transformer denoisers such as U-ViT and DiT unify conditioning by treating image patches, time, and conditions as tokens and applying self-attention over the full sequence. Recent multimodal diffusion Transformers such as MM-DiT (Esser et al., 2024) and UniDiffuser (Bao et al., 2023b) use joint self-attention over concatenated image and text tokens, enabling in-context conditioning without explicit cross-attention.

CFG remains useful for perceptual alignment, but it adds an extra forward pass and can shift the sampling distribution. Guidance-free objectives (Tang et al., 2025) remove the unconditional pass and speed up inference while preserving conditional quality, which can be preferable when faithful conditional modeling is the goal.

3.2. Inpainting, Imputation, and Conditional Sampling

Inpainting and imputation both realize conditional sampling where observed variables remain fixed and the model generates the missing subset. Image diffusion inpainting typically provides a binary mask as an extra channel or additive embedding, and Palette (Saharia et al., 2022) frames colorization, inpainting, uncropping, and restoration under a unified image-to-image diffusion interface. RePaint (Lugmayr et al., 2022) shows that an unconditional denoiser can satisfy the mask constraint by resampling the known region during reverse diffusion, enabling arbitrary masks without mask-specific training.

Tabular imputation follows the same principle by denoising only missing entries while conditioning on observed fields. TabCSDI adapts conditional score-based diffusion to mixed-type tables, and CSDI (Tashiro et al., 2021) demonstrates the same mechanism for time series. Across modalities, a single Transformer backbone can implement conditional sampling by encoding a binary mask as a channel or embedding.

3.3. Any-Subset Conditioning

In tabular settings, any-subset conditioning specifies $C \subset \mathcal{F}$ and $T \subset \mathcal{F}$ over features $\mathcal{F} = \{1, \dots, F\}$, with $C \cap T = \emptyset$ and $C \cup T = \mathcal{F}$. A common formulation uses masks $m_C, m_T \in \{0,1\}^F$ with $m_C + m_T = \mathbf{1}$, and writes

$$x_C^{(0)} = m_C \odot x^{(0)}, \quad x_T^{(0)} = m_T \odot x^{(0)}, \tag{3}$$

The goal is to model $p(x_T^{(0)} \mid x_C^{(0)})$ for any (C, T), as in CSDI and TabCSDI. A common implementation keeps $x_C^{(0)}$ fixed while noising $x_T^{(0)}$. When the missing pattern is unknown, CSDI samples a target ratio $\rho \sim \mathrm{Unif}(0,1)$ and selects that fraction uniformly at random (Tashiro et al., 2021), inducing

$$\begin{aligned} \rho \sim \mathrm{Unif}(0,1), \quad n = \mathrm{round}(\rho F), \\ T \sim \mathrm{Unif}(\mathcal{S}_n), \\ m_T = \mathbf{1}_T, \quad m_C = \mathbf{1} - m_T, \end{aligned} \tag{4}$$

where $\mathcal{S}_n = \{S \subset \mathcal{F} : |S| = n\}$. The partial noising process is then

$$x_i^{(t)} = \begin{cases} x_i^{(0)}, & m_{C,i} = 1, \\ \sqrt{\alpha_t}\, x_i^{(0)} + \sqrt{1-\alpha_t}\, \epsilon_i, & m_{T,i} = 1, \end{cases} \tag{5}$$

The denoiser is conditioned on m_C through a mask channel or embedding, and training minimizes

$$\mathbb{E}_{x^{(0)}, m_C, t, \epsilon}\big[\|m_T \odot (\epsilon - \epsilon_\theta(x^{(t)}, t, m_C))\|^2\big], \tag{6}$$

which encourages the model to handle diverse missingness patterns. The mask distribution p_C should match expected test-time patterns to avoid conditional shift.

4. Our Framework

We introduce QuADiT, a token-based diffusion framework for any-subset conditioning with query-adaptive execution. Our design follows a DiT-style backbone but adapts the input representation and conditioning so that attention excludes non-query features and masked tokens are ignored. The framework targets mixed feature types and can extend to multimodal settings with image features. As shown in Figure 1, a single trained model can perform marginal, joint, and conditional sampling depending on the query specification. Figure 2 summarizes the end-to-end pipeline and the feature-token interface to the diffusion transformer.

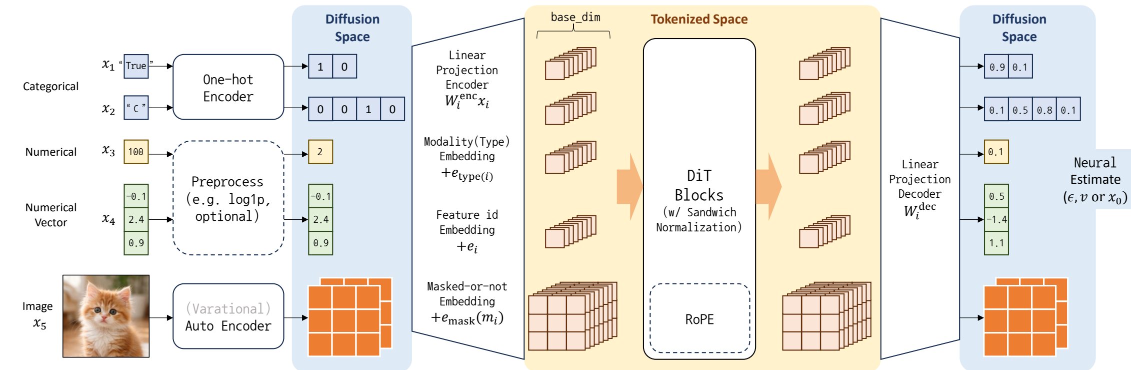

Figure 2. Overview of QuADiT. Each feature is encoded into one (or more for image) token by feature-specific linear projections, then combined with type, feature-id, and mask embeddings before DiT blocks. The decoder maps token outputs back to data space for diffusion prediction, and optional image tokens follow the same interface. See Section 4 for details.

4.1. Feature Tokenization

Let the schema define F features with types in $\{\text{num}, \text{cat}, \text{vector}, \text{image}\}$. Each feature is mapped to one or more tokens of dimension D via a feature-specific linear encoder W_i^{enc}, and the Transformer output is projected back to the data space via a corresponding linear decoder W_i^{dec}. For numerical or vector features, optional preprocessing can be applied depending on the distribution (e.g., $\log(1+x)$). Numerical features are projected from their scalar value, vector features from their raw vector, and image features yield multiple spatial tokens. For categorical features, we follow the multinomial diffusion framework of TabDDPM and argmax flows. A categorical variable $y \in \{1, \dots, C\}$ is encoded as a one-hot vector $z = e_y \in \{0,1\}^C$ and projected via $W_i^{\text{enc}} z$; the decoder outputs logits $\hat{z} = W_i^{\text{dec}} h_i \in \mathbb{R}^C$ which are decoded as $\hat{y} = \arg\max_k \hat{z}_k$.

After projection, we add three learnable embeddings to each token,

$$h_i = W_i^{\text{enc}} x_i + e_{\text{type}(i)} + e_i + e_{\text{m}}(m_i), \tag{7}$$

where $e_{\text{type}(i)}$ is a *type embedding* that provides shared inductive bias among features of the same kind, e_i is a *feature-id embedding* that assigns a unique representation to each column, and $e_{\text{m}}(m_i)$ is a *shared masked-or-not embedding* indexed by $m_i \in \{0,1\}$ to distinguish conditioning features ($m_i = 0$) from to-generate features ($m_i = 1$). For image tokens that require spatial structure, we apply 2D rotary positional embedding (RoPE) (Su et al., 2021) inside the attention mechanism after the embeddings are added; tabular tokens receive no positional encoding since column identity is captured by e_i.

4.2. Query-Adaptive Feature Tokens

Prior any-subset conditioning methods such as CSDI and TabCSDI always feed all F features into the model and apply a mask to select which features to denoise. Even when a query involves only a few variables, the model computes attention over all features and discards outputs not in the target. For example, given features A, B, C, estimating $p(A|B)$ under these methods requires sampling $p(A, C|B)$ and discarding C, so all three features participate in computation even though C is irrelevant to the query. Since self-attention has $O(F^2)$ complexity in the number of features, this fixed-length approach incurs $O(F^2)$ cost per layer regardless of query size.

We address this inefficiency by introducing a query set $Q \subseteq \mathcal{F}$ that explicitly limits which features enter the model. The model tokenizes and attends only to features in Q, while features outside Q are never encoded or processed. In the above example, query-adaptive execution uses only A and B to estimate $p(A|B)$ directly, excluding C entirely. Within Q, a target set $T \subseteq Q$ specifies which features to denoise; the remaining features $Q \setminus T$ serve as clean conditioning context. At inference, we construct per-sample token lists containing only the query-relevant features, reducing attention complexity from $O(F^2)$ to $O(|Q|^2)$ per layer.

We represent these sets with binary masks $m_Q, m_T \in \{0,1\}^F$ satisfying $m_T \le m_Q$ elementwise. During training, we sample m_Q and m_T via a two-stage procedure controlled by hyperparameters $p_{\text{full-Q}}$ and $p_{\text{full-T}}$. We draw a random permutation σ of the F feature indices. With probability $p_{\text{full-Q}}$ we set $m_Q = \mathbf{1}$; otherwise we sample $|Q| \sim \text{Unif}\{1, \dots, F\}$ and set m_Q to the indicator of the first $|Q|$ elements under σ. Given m_Q, with probability $p_{\text{full-T}}$ we set $m_T = m_Q$; otherwise we sample $|T| \sim \text{Unif}\{1, \dots, |Q|\}$ and mark the first $|T|$ elements as targets. The parameter $p_{\text{full-Q}}$ ensures the model still sees full-context examples to learn global dependencies, while $p_{\text{full-T}}$ controls how often the model generates all query features versus a strict subset. Unlike prior any-subset methods such as CSDI and TabCSDI that always condition on the full feature set and vary only the target mask, we additionally

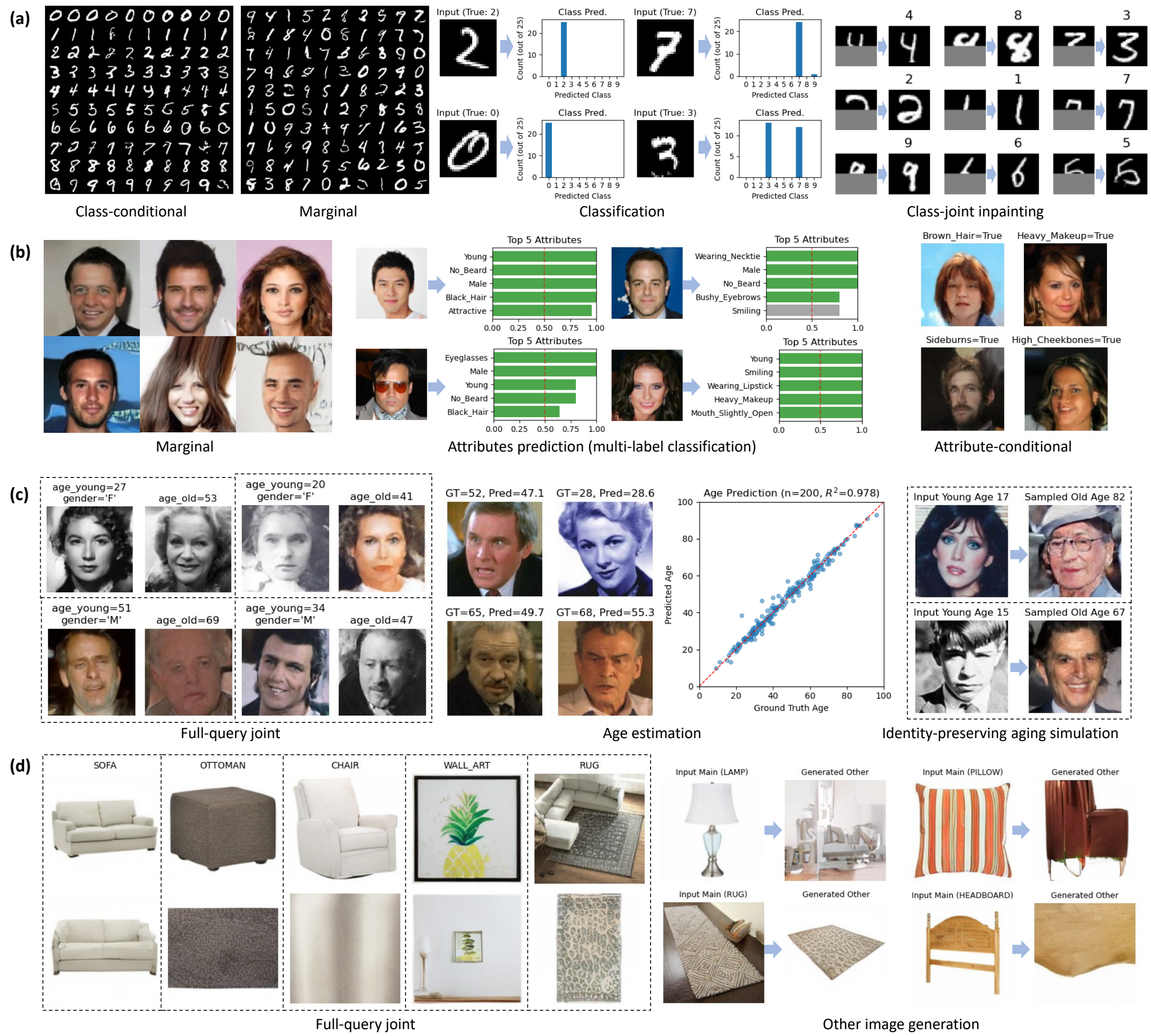

Figure 3. Qualitative results for image-tabular any-subset conditioning. Rows correspond to (a) MNIST (LeCun et al., 1998), (b) CelebA (Liu et al., 2015), (c) AgeDB (Moschoglou et al., 2017), and (d) ABO (Collins et al., 2022); each row is sampled from a single model, illustrating the flexibility of QuADiT. The MNIST row shows class-conditional and marginal sampling, image-to-label prediction with class posteriors, and class-joint inpainting. The CelebA row shows marginal faces, multi-attribute prediction, and attribute-conditioned synthesis. The AgeDB row visualizes joint tuples over (gender, age_young, img_young, age_old, img_old), scatter-based age estimation with example pairs, and identity-preserving aging. The ABO row shows joint product-type-image generation and cross-view completion. Dashed boxes indicate paired samples and blue arrows denote conditioning relationships. Except for a few samples intentionally selected to highlight interesting conditions, most images are non-cherrypicked. See Section 5.1 for experimental details.

vary the query set itself during training. We draw inspiration from classifier-free guidance (Ho & Salimans, 2022), which randomly drops the conditioning around 10–20% of the time to expose the model to varying conditioning regimes; we adopt $p_{\text{full-Q}} = 0.5$ and $p_{\text{full-T}} = 0.25$ as defaults based on this principle.

Let M_T denote the element-level expansion of m_T over the data dimensions. We construct the partially noised input as

$$\begin{aligned} x^{(t)} &= M_T \odot \tilde{x}^{(t)} + (1 - M_T) \odot x^{(0)}, \\ \tilde{x}^{(t)} &= \sqrt{\alpha_t}\, x^{(0)} + \sqrt{1-\alpha_t}\, \epsilon. \end{aligned} \tag{8}$$

Conditioning features thus enter the model without added noise, providing clean context for denoising the target features. We pass $\bar{m}_Q = 1 - m_Q$ as a key padding mask so that attention excludes non-query tokens. At inference, we follow the repaint-style approach where conditioning features are replaced with their clean values $x^{(0)}$ after each reverse step.

Figure 4 empirically confirms that inference cost scales with query size. When only a subset of features is queried, query-adaptive execution avoids unnecessary computation on irrelevant tokens.

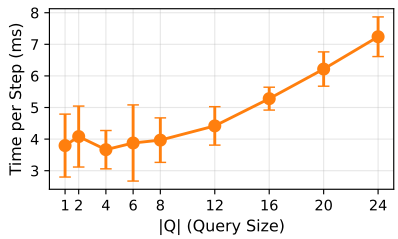

Figure 4. Per-step inference time versus query size $|Q|$ on the Default dataset (batch size 256, NVIDIA RTX 3090). Time grows with query size, confirming that computation scales with $|Q|$ rather than F.

4.3. Implementation Details

Noise Schedule. We use the cosine schedule (Nichol & Dhariwal, 2021) that defines $\alpha_t = \cos^2\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)$ with a small offset $s = 0.008$. Compared to the linear schedule, the cosine schedule reduces information destruction at early timesteps and yields smoother training dynamics.

Prediction Target. The denoiser can predict the noise ϵ, the clean sample $x^{(0)}$, or the velocity $v = \sqrt{\alpha_t}\,\epsilon - \sqrt{1-\alpha_t}\,x^{(0)}$ introduced by Salimans and Ho (Salimans & Ho, 2022). Given model output $\hat{v}$, the other quantities are recovered as

$$\begin{aligned} \hat{x}^{(0)} &= \sqrt{\alpha_t}\,x^{(t)} - \sqrt{1-\alpha_t}\,\hat{v}, \\ \hat{\epsilon} &= \sqrt{1-\alpha_t}\,x^{(t)} + \sqrt{\alpha_t}\,\hat{v}. \end{aligned} \tag{9}$$

The v-prediction objective interpolates between ϵ- and $x^{(0)}$-prediction and provides more balanced gradients across timesteps. We adopt v-prediction as our default.

Sandwich Normalization. Recent diffusion Transformers such as Next-DiT (Zhuo et al., 2024) adopt sandwich layer normalization to stabilize training and inference. We apply root mean square normalization (RMSNorm) before and after the attention and feedforward blocks, combined with adaptive modulation (adaLN-Zero) for time conditioning. This stabilizes gradients and accelerates convergence, which we confirm in our ablation.

Training Objective. Let $\mathcal{L}_i = \|\hat{v}_i - v_i\|^2$ denote the per-element squared error. Following Hang et al. (Hang et al., 2023), we weight each sample by $\min(\mathrm{SNR}_t, \gamma)/(\mathrm{SNR}_t + 1)$ with $\gamma = 5$, where the signal-to-noise ratio is $\mathrm{SNR}_t = \alpha_t/(1-\alpha_t)$. This min-SNR weighting balances conflicting gradients across timesteps. To handle heterogeneous feature dimensions, we further weight each element by $1/\sqrt{d_i}$ where d_i is the data dimension of feature i, so that high-dimensional features do not dominate the loss. The final loss averages over target elements within each sample first, then over samples.

Training Configuration. We largely follow the training configuration of DiT. We use the AdamW optimizer with a learning rate of 10^{-4}, gradient clipping with a maximum norm of 1.0, and T=1000 diffusion steps. Exponential moving average (EMA) of model weights with decay rate 0.999 is maintained throughout training and used for evaluation.

5. Experiments

We focus on demonstrating the generality of QuADiT as a unified framework for any-subset conditional sampling across tabular and image-tabular modalities, and on identifying robust design choices through ablation rather than chasing task-specific benchmarks.

5.1. Qualitative Results on Image-Tabular Generation

We first demonstrate that QuADiT supports multimodal any-subset conditional sampling by evaluating image-tabular generation. We choose some well-known vision datasets with rich tabular attributes to highlight the intended setting for image-tabular conditioning.

Datasets. We use MNIST (LeCun et al., 1998) as a joint distribution over a digit category and an image. CelebA (Liu et al., 2015) provides 40 binary attributes paired with each face image. AgeDB (Moschoglou et al., 2017) includes gender and faces across many ages for each identity, so we sample an identity and two ages to form tuples $(\mathsf{gender}, \mathsf{age\_young}, \mathsf{img\_young}, \mathsf{age\_old}, \mathsf{img\_old})$. For product imagery, we use ABO (Collins et al., 2022) and construct $(\mathsf{product\_type}, \mathsf{img\_main}, \mathsf{img\_other})$ pairs, where the other images typically capture close-ups of texture, measurements, or alternate viewpoints.

Image encoding. We test both pixel-space and latent-space representations. For MNIST, we treat each image row as a vector feature and omit any positional encoding, effectively using a pure tabular encoding of pixels. For CelebA, AgeDB, and ABO, we resize images to 128^2 and encode them with pretrained variational autoencoders (VAEs) in the latent space, following the latent diffusion practice. We use the Qwen-Image VAE for face datasets (Wu et al., 2025) and the DC-AE f32c32 autoencoder from Sana (Chen et al., 2024; Xie et al., 2024) for ABO. The resulting latents are treated as image features and receive 2D RoPE in the attention blocks.

Model configuration. MNIST uses a compact backbone with hidden size 256, depth 4, and 4 attention heads, resulting in about 6M parameters. For the other datasets, we adopt a DiT-S configuration with hidden size 384, depth 12, and 6 heads, totaling about 33M parameters. Training uses AdamW with learning rate 10^{-4}, batch size

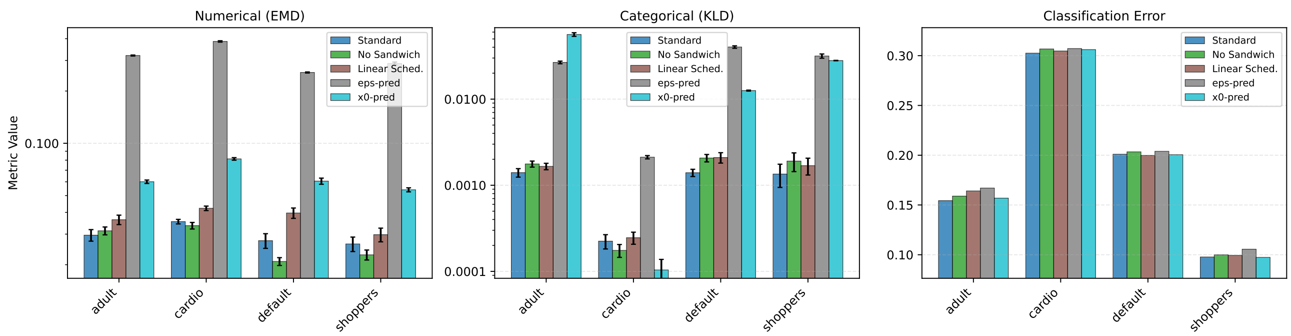

Figure 5. Ablation study on key design choices: Sandwich Normalization, noise schedule, and prediction target (v, $x^{(0)}$, ϵ) across four tabular benchmarks. We report marginal fitting (Earth Mover's Distance (EMD) for numerical, Kullback-Leibler divergence (KLD) for categorical features) and binary classification errors (lower is better). Error bars indicate the standard deviation over five runs and are negligible for the classification metrics.

256, and T=1000 diffusion steps, extended to 1M steps for image-containing datasets. For these datasets, we set $p_{\text{full-Q}} = p_{\text{full-T}} = 0.7$, which ensures that images are included as targets with probability $0.7^2 \approx 0.5$, allowing the model to observe and learn from images more frequently for faster convergence. This setup allows any-subset queries such as attribute-conditioned image generation, attribute completion from images, and cross-view image synthesis within a single model.

Results. Figure 3 provides dataset-specific qualitative evidence of any-subset conditioning across tabular and image modalities. Rows correspond to datasets and show marginal, joint, and conditional queries with paired samples and conditioning arrows. The MNIST row shows class-conditional and marginal sampling, image-to-label prediction with class posteriors, and class-joint inpainting; the CelebA row shows marginal faces, multi-attribute prediction, and attribute-conditioned synthesis. The AgeDB row illustrates joint tuples over $(\mathsf{gender}, \mathsf{age\_young}, \mathsf{img\_young}, \mathsf{age\_old}, \mathsf{img\_old})$, scatter-based age estimation with example pairs, and identity-preserving aging, while the ABO row shows joint product-type-image generation and cross-view completion. Repeated sampling yields empirical distributions for tabular attributes or ages rather than single point estimates. Additional joint samples for CelebA, AgeDB, and ABO are provided in Figures 10 to 12.

5.2. Design Choice Ablation

We next ablate key design choices on four tabular benchmarks with mixed numerical and categorical variables and binary targets, namely Adult (Becker & Kohavi, 1996), Cardio (Halder, 2020), Default (Yeh, 2009), and Online Shoppers (Sakar & Kastro, 2018), spanning demographic, medical, financial, and e-commerce domains. Preprocessing can be chosen as needed for the dataset. In our experiments we

Table 1. Win rates across 12 dataset-metric pairs.

Ablation	Option	Wins
Sandwich Normalization	**On**	**8/12**
	Off	4/12
Noise Schedule	**Cosine**	**11/12**
	Linear	1/12
Prediction Target	**v**	**9/12**
	$x^{(0)}$	3/12
	ϵ	0/12

apply $\log(1+x)$ for heavy-tailed numerical columns; for more unusual distributions, one can also consider quantile transformer as in TabDDPM. Numerical-feature metrics are computed in the tokenized diffusion space rather than the raw data space, so these preprocessing choices are not critical to the ablation results. We train all models with batch size 256 for 200k steps. We use a compact backbone smaller than DiT-S, with hidden size 256, depth 4, and 4 attention heads (~6M parameters), the same configuration used for the MNIST experiments above.

We vary three design choices around the standard configuration: sandwich normalization, noise schedule, and prediction target, while keeping the backbone and training setup fixed. For marginal fidelity, we sample 50,000 joint rows with 50-step DDIM (Denoising Diffusion Implicit Models) (Song et al., 2021a) and report EMD for numerical features and KLD for categorical features. For conditional utility, we generate targets five times per sample and use majority vote to measure binary classification error. The full metric profiles are shown in Figure 5. Per-feature histograms that compare data and unconditional samples for each benchmark appear in Appendix Additional Results on Tabular Benchmarks in Figure 9.

Because metric scales differ across datasets and feature types, we summarize the ablations with win rates computed

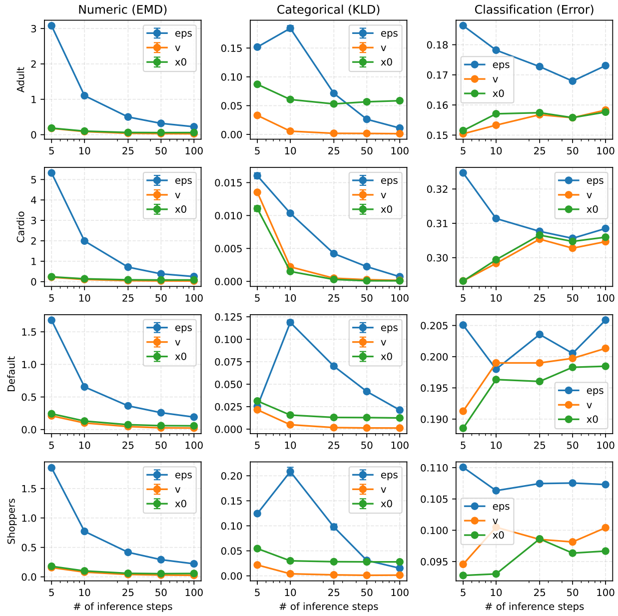

Figure 6. Effect of DDIM inference steps on marginal fidelity and conditional utility across four tabular benchmarks. We sweep 5, 10, 25, 50, and 100 steps and report EMD (numerical), KLD (categorical), and classification error (lower is better; error bars show std over 5 runs).

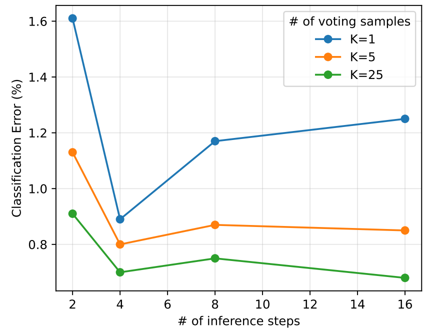

Figure 7. MNIST classification error when conditioning on the image and sampling the class token with DDIM for 2, 4, 8, and 16 steps. We report majority-vote error for $K \in \{1, 5, 25\}$ label samples, and lower is better.

over 12 dataset-metric pairs (four datasets and three metrics), shown as wins/12. Table 1 shows that the cosine schedule dominates in 11 of 12 cases, and v-prediction wins 9 of 12 cases while ϵ-prediction never wins. Sandwich normalization improves classification error and KLD in most datasets but slightly hurts EMD on three datasets, yielding a smaller yet still favorable win rate. We therefore adopt cosine, v-prediction, and sandwich normalization as the standard setting while noting that sandwich can be relaxed if numerical marginal fidelity is the sole priority.

5.3. Number of Inference Steps

DDIM accelerates sampling by reducing inference steps, but too few steps can harm fidelity. To quantify this trade-off, we sweep 5, 10, 25, 50, and 100 steps for each prediction target on the tabular benchmarks. Figure 6 shows that v-prediction generally achieves the best or tied performance on marginal fidelity, with $x^{(0)}$ close behind and ϵ lagging. Most metrics flatten by 50 steps, consistent with DDIM practice in image diffusion.

We observe two counterintuitive patterns. First, ϵ-prediction yields a sudden improvement in KLD at 5 steps even though other metrics deteriorate. Second, classification error for v and $x^{(0)}$ slightly improves at lower step counts.

To further investigate the step-error relationship for categorical prediction, we measure MNIST classification by conditioning on the image and sampling the class token with DDIM, then taking a majority vote among K samples, which acts as a Monte Carlo estimate of $p(y|x)$ and is conceptually related to Diffusion Classifier (Li et al., 2023). Figure 7 sweeps 2, 4, 8, and 16 steps with $K \in \{1, 5, 25\}$. Increasing K consistently improves accuracy, and fewer steps can slightly reduce error. This pattern mirrors the tabular classification results (Figure 8 in the Appendix), where all four benchmarks show that fewer inference steps improve classification error when combined with majority voting. Both effects involve categorical features and hint at more efficient sampling strategies, but we leave a concrete explanation for future work.

6. Conclusion

We introduced QuADiT, a diffusion Transformer that represents mixed-type tables and images as feature tokens and supports any-subset conditional sampling via query masks and query-adaptive execution. Across tabular benchmarks, ablations on schedule, prediction target, and normalization identify a strong standard configuration, and inference-step sweeps show that 50-step DDIM is largely sufficient.

Beyond tabular synthesis, the same backbone models image-tabular joint distributions and supports diverse tasks in a single model, including attribute-conditioned generation, attribute completion, and cross-view synthesis. Our goal is not to chase state-of-the-art numbers on a single benchmark, but to establish a unified and extensible framework that generalizes across feature types and modalities. We consistently observe that very small inference steps can slightly improve categorical label classification, yet a clear explanation is still missing. Future work includes more efficient categorical sampling, theoretical analysis of query-adaptive diffusion, and larger-scale multimodal benchmarks.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Additional Results on Tabular Benchmarks

We provide supplementary results from the tabular benchmark experiments described in Section 5.2. Figure 8 extends the inference-step analysis to all four benchmarks, and Figure 9 visualizes per-feature marginal distributions comparing real data and model samples.

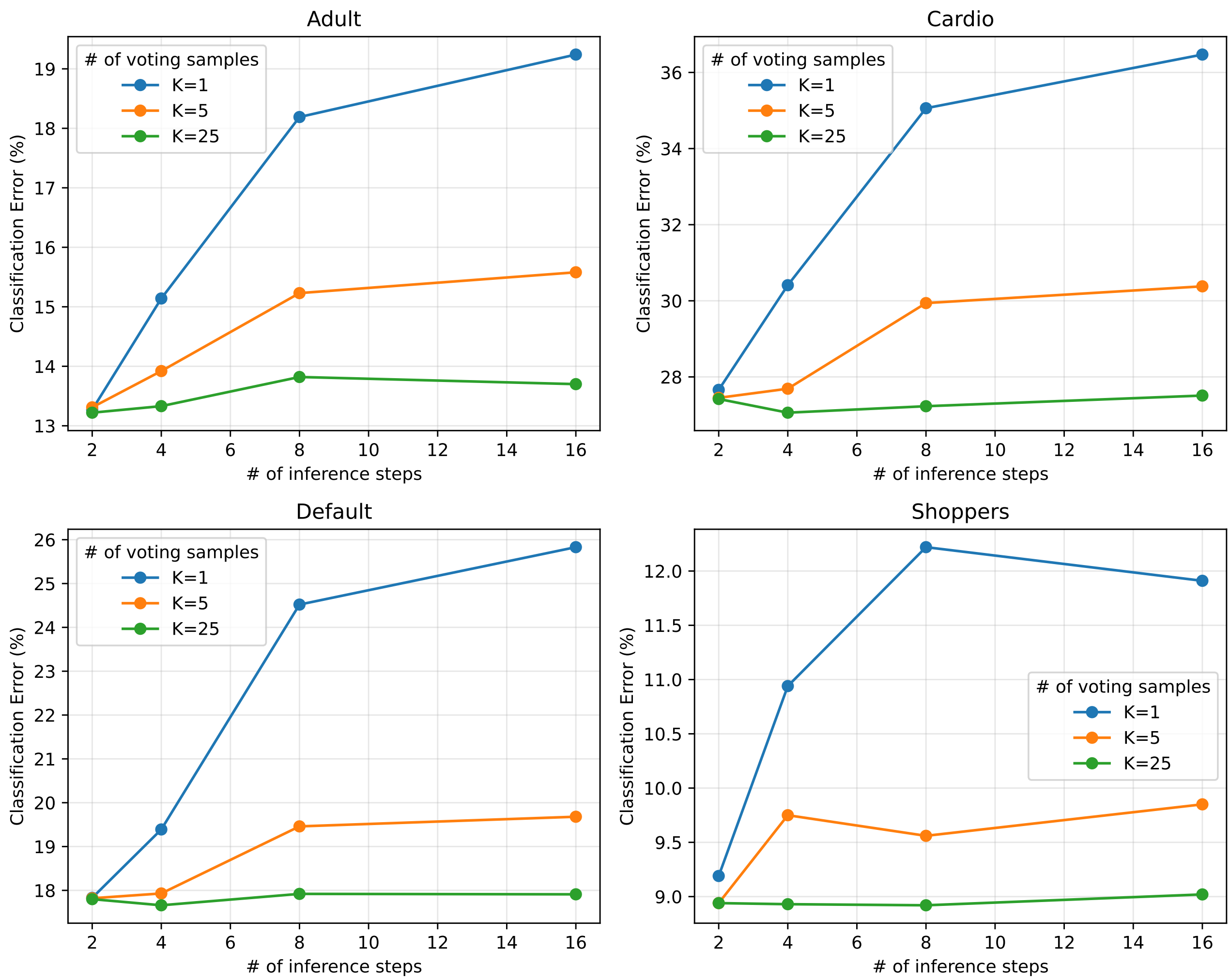
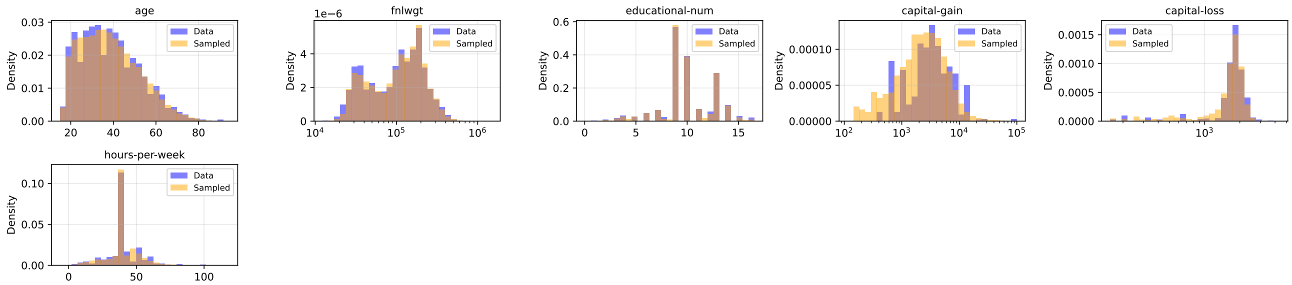
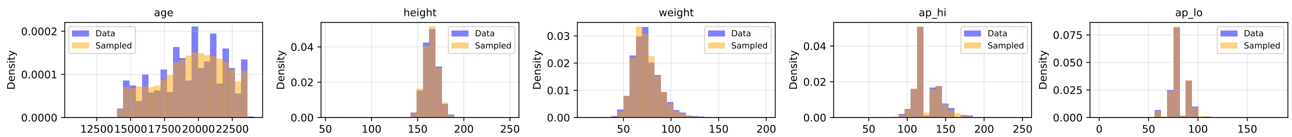


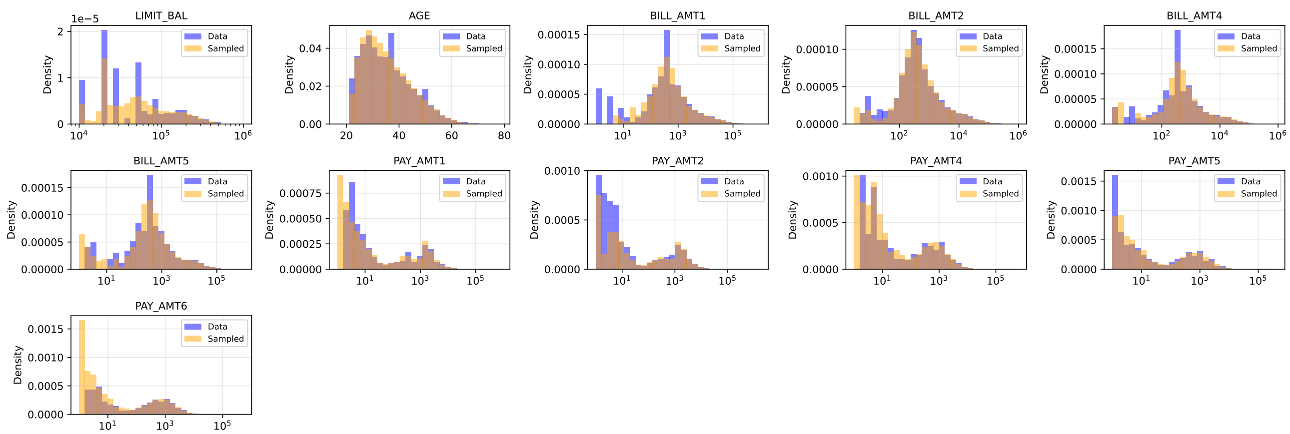
Figure 8. Classification error on four tabular benchmarks when varying the number of DDIM inference steps (2, 4, 8, 16) and the number of voting samples $K \in \{1, 5, 25\}$. Consistent with the MNIST results in Figure 7, fewer inference steps can reduce classification error, and increasing K consistently improves accuracy across all datasets. Unlike MNIST, these errors are measured on each dataset's training set.



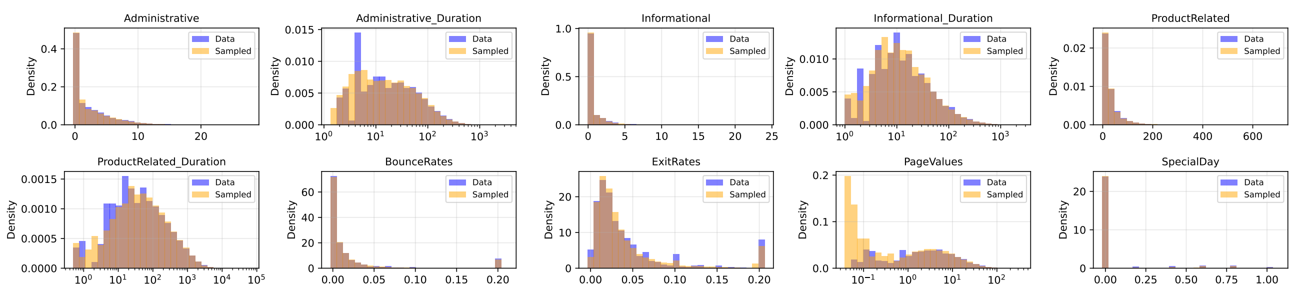
(a) Adult



(b) Cardio



(c) Default



(d) Online Shoppers

Figure 9. Per-feature histograms comparing data and sampled marginals on four tabular benchmarks. (n=30,000)

B. Additional Image-Tabular Joint Samples

We present additional joint samples from the image-tabular experiments in Section 5.1. These figures show unconditional joint samples drawn from each trained model, illustrating that the learned distributions capture coherent relationships between images and their associated tabular attributes.

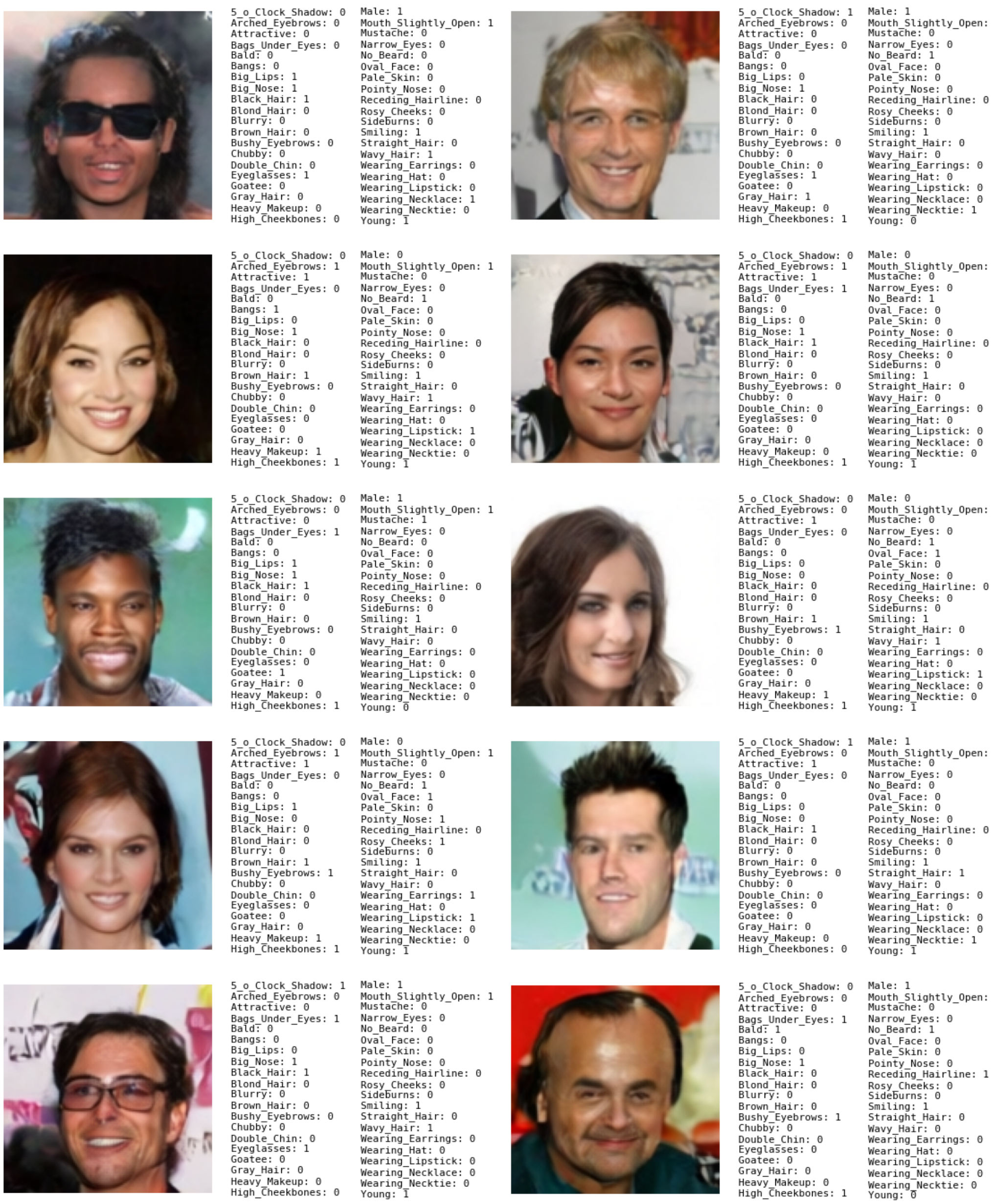


Figure 10. Joint samples from the CelebA model. Each sample shows a face image alongside its 40 binary attributes. The model learns the joint distribution over images and attributes, enabling both marginal sampling and attribute-conditioned generation.

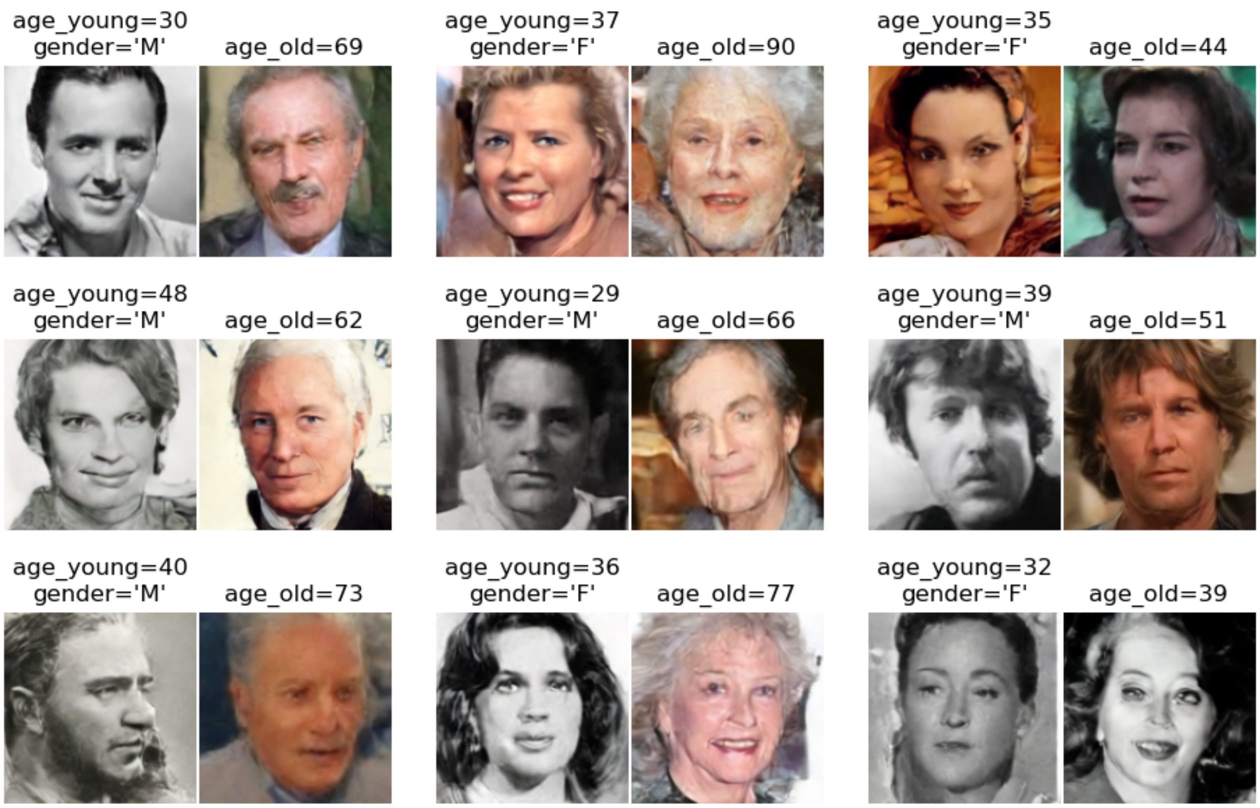


Figure 11. Joint samples from the AgeDB model over $(\mathsf{gender}, \mathsf{age\_young}, \mathsf{img\_young}, \mathsf{age\_old}, \mathsf{img\_old})$ tuples. Each row shows a sampled identity with two age snapshots, demonstrating the model's ability to capture identity-preserving age variations.

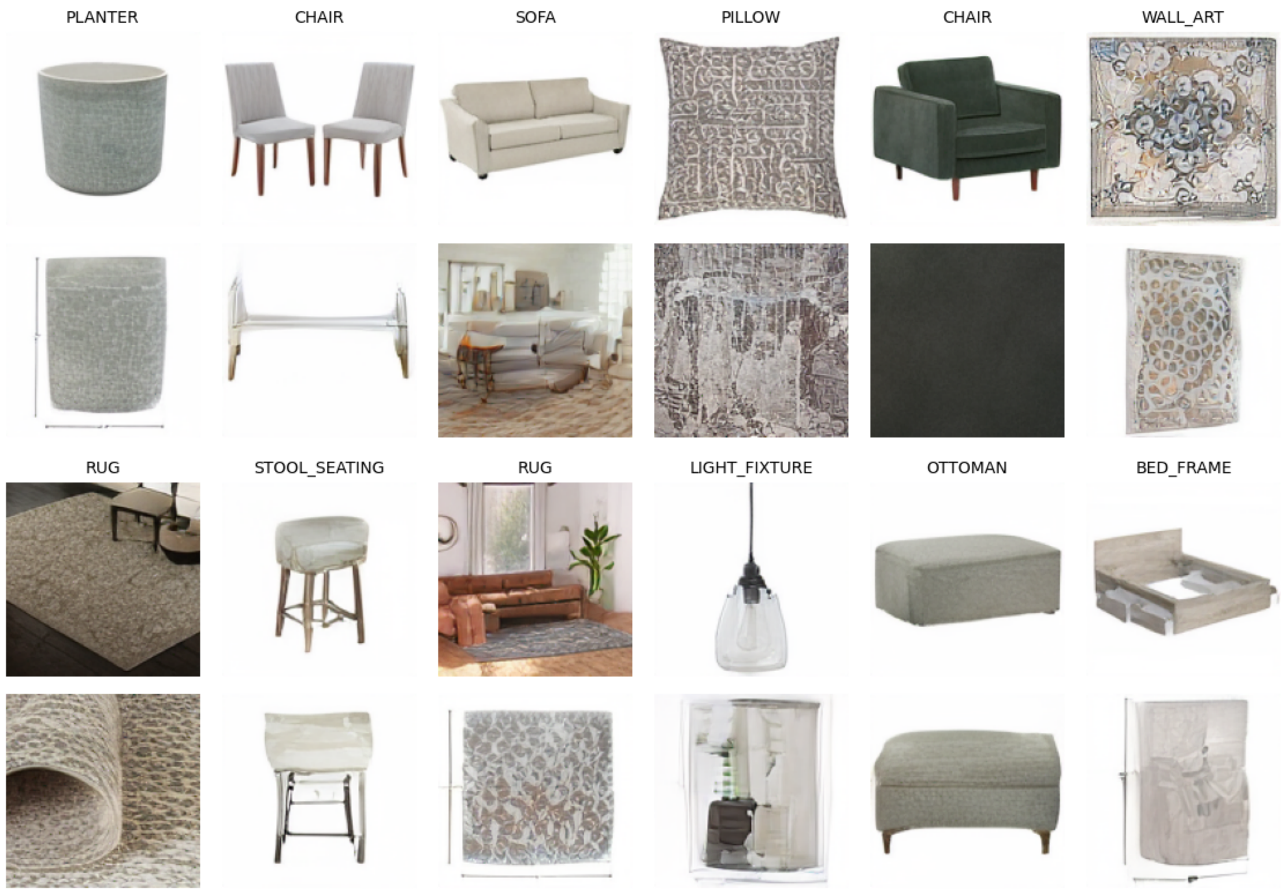


Figure 12. Joint samples from the ABO model over (product_type, img_main, img_other) tuples. The model generates coherent product images with matching category labels, where main and other views maintain visual consistency.